

Effects of ectomycorrhizas and vesicular–arbuscular mycorrhizas, alone or in competition, on root colonization and growth of *Eucalyptus globulus* and *E. urophylla*

Y. L. CHEN¹, M. C. BRUNDRETT² AND B. DELL^{3*}

¹*The Research Institute of Tropical Forestry, Chinese Academy of Forestry, Longdong, Guangzhou, 510520, P. R. of China*

²*CSIRO Forestry and Forest Products, CSIRO Centre for Mediterranean Agricultural Research, Private Bag PO Wembley, WA 6014, Australia*

³*School of Biological Sciences and Biotechnology, Murdoch University, Perth WA 6150, Australia*

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SUMMARY

Eucalyptus species are considered to have ectomycorrhizas (ECM), but many also have vesicular–arbuscular mycorrhizas (VAM) and their relative importance is unclear. Interactions between ECM and VAM fungi colonizing roots of *Eucalyptus* species were examined in a glasshouse experiment. This experiment investigated competition between these two types of fungi and compared benefits provided to *Eucalyptus globulus* and *E. urophylla*. *Eucalyptus* seedlings were inoculated with spores of the ECM fungus *Laccaria lateritia* and/or pot-culture soil for a VAM fungus (species of *Glomus*, *Acaulospora* or *Scutellospora*). Initial inoculum levels were important, as VAM fungi became established much more rapidly than *Laccaria*. In plants with both types of mycorrhizas, *Laccaria* mycorrhizas substantially increased after 2 or 3 months and the proportion of roots with VAM declined. However, the proportion of roots with VAM also decreased significantly in plants without ECM after 2 months. Thus, the relative susceptibility of eucalypt roots to these mycorrhizal associations changed. Substantial growth responses to mycorrhizal inoculation occurred when a low concentration of phosphorus fertilization was used (5 mg kg⁻¹), but not at a higher concentration (10 mg kg⁻¹). Treatments where *Laccaria* was applied, alone or in combination with a VAM fungus, resulted in the largest growth increases relative to nonmycorrhizal plants. An *Acaulospora* isolate was the most effective VAM fungus for *E. urophylla* although other VAM fungi also increased growth relative to the control. VAM fungi alone had little effect on *E. globulus* growth, but plants inoculated with both ECM and VAM fungi were larger than plants only inoculated with *Laccaria*. ECM fungi had a major impact on root system form by reducing the proportion of fine roots (specific root length) relative to nonmycorrhizal plants or those with VAM. Changes in root colonization patterns over time must be considered in studies of ECM/VAM interactions, as there can be substantial changes in the relative importance of the two associations.

Key words: ectomycorrhizas, vesicular–arbuscular mycorrhizas, *Eucalyptus*, inoculation, competition, growth responses.

INTRODUCTION

Field surveys have found that plants with mycorrhizas predominate in most natural ecosystems, including those in Australia (Brundrett, 1991).

Vesicular–arbuscular mycorrhizas (also called arbuscular mycorrhizas and VAM) are the most prevalent of these associations in Australian plant communities, but trees and plants with ectomycorrhizal associations (ECM) and nonmycorrhizal plants are also important (Brundrett, 1991; Brundrett & Abbott, 1991). In Australia, plants with ECM associations usually also have some VAM in

*Author for correspondence (tel +61 89 360 2875; +61 89 360 6303; dell@central.murdoch.edu.au).

their roots, but little is known about the relative importance of these two associations. Australian plants reported to have dual VAM/ECM associations include major species used in plantation forestry belonging to the genera *Casuarina*, *Allocasuarina* (Casuarinaceae), *Eucalyptus*, *Melaleuca* (Myrtaceae) and *Acacia* (Mimosaceae) (Brundrett *et al.*, 1996). Plants with dual ECM/VAM associations are less often reported from other parts of the world (Brundrett, 1991), but there are exceptions such as *Alnus*, *Populus*, *Salix* and *Uapaca* (Lodge & Wentworth, 1990; Zhong, 1995; Moyersoen & Fitter, 1999).

Eucalypts have an important role for wood production in plantations in Australia (National Forest Inventory, 1997) and in Asia (Midgley & Pinyopusarerk, 1996). The success of these species results from their capacity to grow on soils with limited phosphorus and nitrogen (Specht, 1996). *Eucalyptus* can form both ECM and VAM in the same root systems both in plantation soils and under controlled conditions (Lapeyrie & Chilvers, 1985; Brundrett *et al.*, 1996; Oliveira *et al.*, 1997). However, only the ECM associations have been considered in studies of *Eucalyptus* root structure, or physiology (e.g. Chilvers & Pryor, 1965; Hilbert & Martin, 1988).

Ectomycorrhizas have been shown to provide substantial growth responses when eucalypts are grown in infertile soils and are considered to be important to tree productivity in plantations (Grove & Malajczuk, 1994; Malajczuk *et al.*, 1994; Brundrett *et al.*, 1996; Gong *et al.*, 1997). Glasshouse experiments comparing the benefits provided to eucalypts by ECM and VAM associations have shown a greater benefit from ECM (Jones *et al.*, 1998). Inoculation with VAM fungi alone substantially promoted the growth of eucalypts in some cases (Lapeyrie & Chilvers, 1985; Jasper & Davy, 1993), but not in others (Adjoud *et al.*, 1996). A succession from VAM to ECM associations in eucalypt roots can occur as trees age in field soils (Gardner & Malajczuk, 1988; Bellei *et al.*, 1992; Oliveira *et al.*, 1997) and in the glasshouse (Lapeyrie & Chilvers, 1985; Chen *et al.*, 1998). Further research is required to determine the relative importance of ECM and VAM associations for eucalypts and how these relationships change with time and with different host or fungus genotypes. Such studies should also be relevant to other important tree genera with dual associations such as *Acacia* and *Casuarina*.

The purpose of this study was to investigate interactions between ECM and VAM associations of two *Eucalyptus* species (*E. globulus* and *E. urophylla*) by: (1) examining the relative susceptibility of eucalypt seedling roots to these fungi, when inoculated alone or together, (2) comparing their effects on plant yield and root system form, (3) comparing

mycorrhizal responses at two soil phosphorus concentrations and (4) following changes in these interactions over the first 4 months of growth.

MATERIALS AND METHODS

A complete randomized block design with 10 fungal treatments (two ECM treatments $\times$ five VAM treatments), two tree species and two P concentrations was used for this experiment. Seedlings were inoculated with spores of the ECM fungus *Laccaria*, or were grown without ECM. For the five VAM treatments, seedlings were inoculated with pot-culture soil containing an isolate of either *Glomus*, *Acaulospora* or *Scutellospora*, or air dried field soil, or they were grown without VAM. For each treatment, there were four replicate pots with two plants in each pot. Fungal treatments were applied at the same time that seedlings were planted.

Eucalyptus globulus Labill. and *E. urophylla* S. T. Blake were chosen for their importance in temperate and tropical plantation forestry, respectively. Seeds were surface sterilized by shaking in a 1.2% solution of NaOCl for 10 min (*E. globulus*), or 15 min with 0.25% NaOCl (*E. urophylla*). After 4 rinses in sterile water, 20 seeds were transferred to each 90-mm Petri dish with 20 ml of germination medium (500 μ M CaSO₄, 3 μ M H₃BO₃, 0.7% Agar). Plates were incubated in darkness at 25°C for 8 d and once germination had started, placed in the light for 2 d before planting.

The ECM fungus *Laccaria lateritia* Malençon is one of the most common species in young West Australian eucalypt plantations (Bougher & Syme, 1998; Lu *et al.*, 1999). This species has been shown to form ECM associations from spore inoculum in glasshouse experiments using the same host plants, soil conditions, inoculum formulation and application method used in the current experiment (Lu *et al.*, 1998). Mature sporocarps of *L. lateritia* were collected from young eucalypt plantations in southwestern Australia, dried at 40°C and stored at room temperature. They were crushed over a 1 mm mesh sieve to make a mixture of basidiospores and fruit body fragments, which was suspended in deionized water (Lu *et al.*, 1998). This suspension was applied near the base of seedlings at 10 ml (*c.* 10⁶ basidiospores) per seedling.

For VAM fungi, mixed soil from dried clover pot cultures (including sandy soil, root fragments and spores) of three VAM fungal isolates, *Glomus invermaium* Hall (WUM 10), *Acaulospora laevis* Gerdemann & Trappe (WUM 11-4) and *Scutellospora calospora* (Nicol. & Gerd.) Walkers & Sanders (WUM 12-2) was used as inoculum. These isolates have been used in a comprehensive series of experiments which demonstrated growth responses in a number of host plants as well as substantial differences in fungal biology (Abbott *et al.*, 1992).

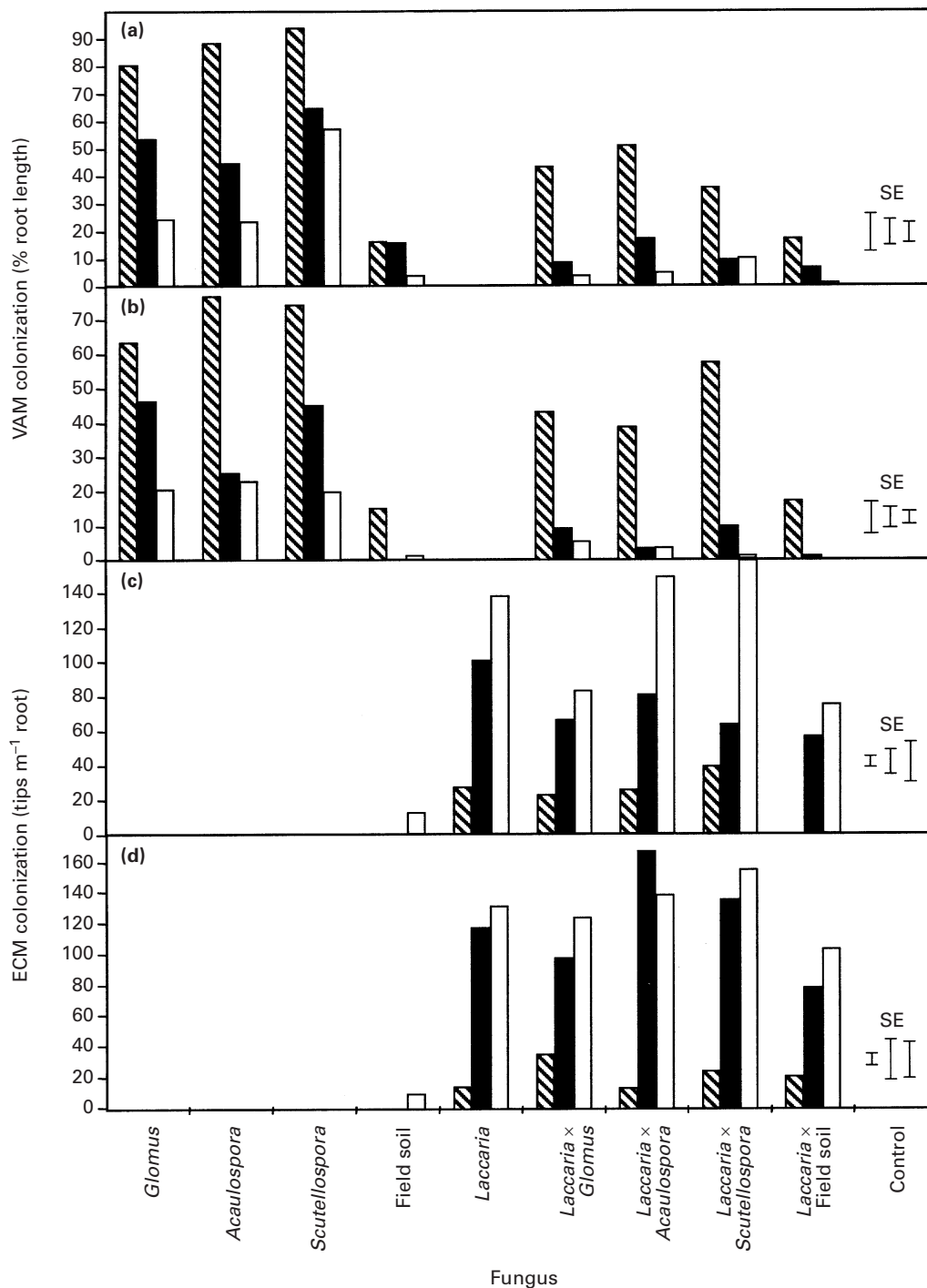


Fig. 1. Mycorrhizal colonization by VAM (a,b) and ECM (c,d) fungi in *Eucalyptus globulus* roots at 8 (hatched columns), 12 (closed columns) and 16 wk old (open columns). Results for two phosphorus concentrations are shown; 5 mg P kg⁻¹ soil (a,c) and 10 mg P kg⁻¹ soil (b,d). Bars are two standard errors of the group mean for each sample time.

Mixed field soil (sandy loam) collected from a clover-based pasture just before establishment of a *Eucalyptus globulus* plantation was used as an additional inoculation treatment. This soil was mixed, sieved (2 mm mesh) and air dried in a glasshouse before use. Pot culture material or field soil (45 g) was layered at a depth of 5 cm in pots before seedlings were planted.

Soil preparation and fertilizer additions were

based on earlier glasshouse trials where responses of *Eucalyptus* spp. to mycorrhizal inoculation were quantified (Bougher *et al.*, 1990; Burgess *et al.*, 1994). The soil used was a Karrakatta Yellow Sand from the Spearwood Dune System north of Perth, Western Australia (soil/water 2:1 slurry, pH 5.4; Bray extractable P < 2 mg P kg⁻¹). Moistened soil was steam sterilized twice at 90°C for 1 h, then oven dried at 70°C, and sieved through a 2.0 mm mesh

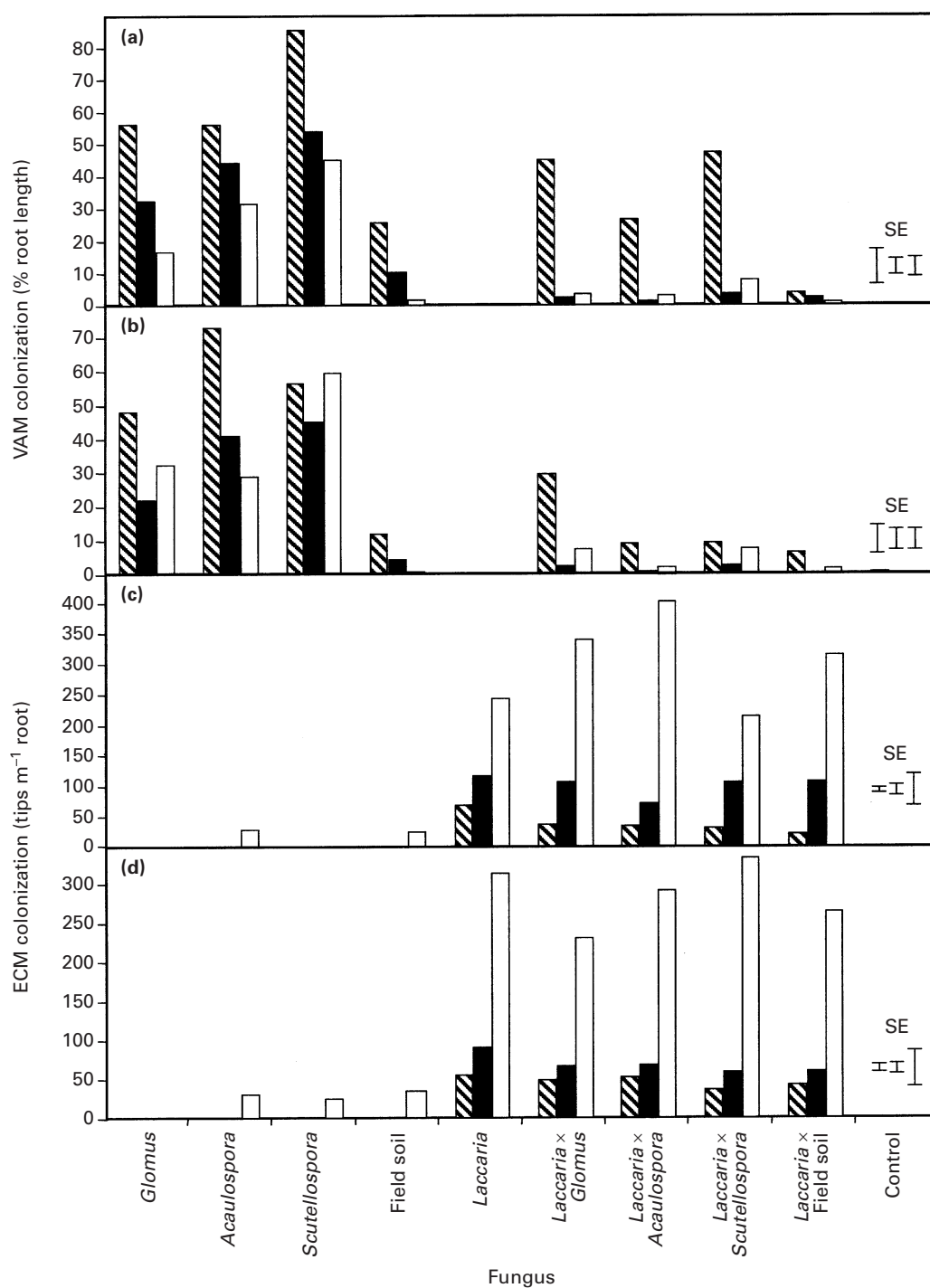


Fig. 2. Mycorrhizal colonization by VAM (a,b) and ECM (c,d) fungi in *Eucalyptus urophylla* roots at 8 (hatched columns), 12 (closed columns) and 16 wk old (open columns). Results for two phosphorus concentrations are shown; 5 mg P kg⁻¹ soil (a,c) and 10 mg P kg⁻¹ soil (b,d). Bars are two standard errors of the group mean for each sample time.

screen. Plastic pots (2-l volume, 14-cm top diameter) were lined with plastic bags and filled with 2 kg of soil. Nutrient solutions, other than P, were applied at concentrations known to be optimum for eucalypt growth in this soil (Burgess *et al.*, 1994, Brundrett *et al.*, 1996). These concentrations were (mg kg⁻¹ soil): CaSO₄·2H₂O, 8.0; K₂SO₄, 48; MgSO₄·7H₂O, 3.2; MnSO₄·4H₂O, 4.4; CuSO₄·5H₂O, 2.3; ZnSO₄·7H₂O, 2.0; CoCl₂·6H₂O, 0.081; (NH₄)₆-

Mo₇O₂₄·4H₂O, 0.24; Na₂B₄O₇·10H₂O, 0.12. Solutions containing these compounds were applied to the surface of soil in pots and allowed to dry in a glasshouse for 72 h, before phosphorus as Ca(H₂PO₄)₂·H₂O in powdered form was added to give a concentration of 5 or 10 mg P kg⁻¹. The soil was then thoroughly mixed by shaking. These P concentrations correspond to points on the lower and upper part of the response curve for these

Table 1. Analysis of the effect of main factors and interactions using the data for growth of two *Eucalyptus* species and the amount of ECM and VAM in their roots at three sampling times

E	Week	P	Plant growth (height)					ECM colonization					VAM colonization				
			ECM	VAM	ECM × VAM	<i>P</i>	Time	ECM	VAM	ECM × VAM	<i>P</i>	Time	ECM	VAM	ECM × VAM	<i>P</i>	Time
1	8	1	0.032	0.358	0.085	0.002	0.000	0.000	0.107	0.107	0.930	0.000	0.066	0.025	0.000	0.650	0.000
		2	0.129	0.814	0.454			0.000	0.258	0.258			0.125	0.006	0.002		
	12	1	0.012	0.411	0.071	0.235		0.000	0.446	0.446	0.500		0.001	0.000	0.236	0.193	
		2	0.430	0.045	0.203			0.000	0.350	0.350			0.072	0.119	0.000		
	16	1	0.000	0.008	0.369	0.198		0.000	0.300	0.142	0.386		0.034	0.093	0.002	0.078	
		2	0.005	0.220	0.469			0.000	0.773	0.087			0.096	0.090	0.000		
2	8	1	0.120	0.863	0.006	0.000	0.000	0.007	0.500	0.023	0.321	0.000	0.048	0.297	0.000	0.112	0.003
		2	0.003	0.751	0.125			0.000	0.918	0.918			0.045	0.286	0.000		
	12	1	0.008	0.522	0.076	0.246		0.000	0.428	0.428	0.500		0.052	0.076	0.002	0.258	
		2	0.025	0.868	0.107			0.000	0.822	0.822			0.034	0.087	0.009		
	16	1	0.023	0.584	0.021	0.394		0.001	0.288	0.035	0.292		0.088	0.188	0.007	0.314	
		2	0.833	0.801	0.015			0.034	0.800	0.001			0.109	0.153	0.000		

Only probability values are shown from separate ANOVAs performed for each combination of eucalypt (E), sample time (week) and phosphorus (P) concentration. The overall significance of changes over time were analysed by repeated measures ANOVA for each *Eucalyptus* species.

E1, *E. globulus*; E2, *E. ourophylla*; P1, P2, 5 and 10 mg P kg⁻¹, respectively; week, time since planting and inoculation of seedlings; ECM colonization, tips m⁻¹ root; VAM colonization, arcsine transformed % root length. The most conservative error term was used to calculate *P* values if interactions were significant.

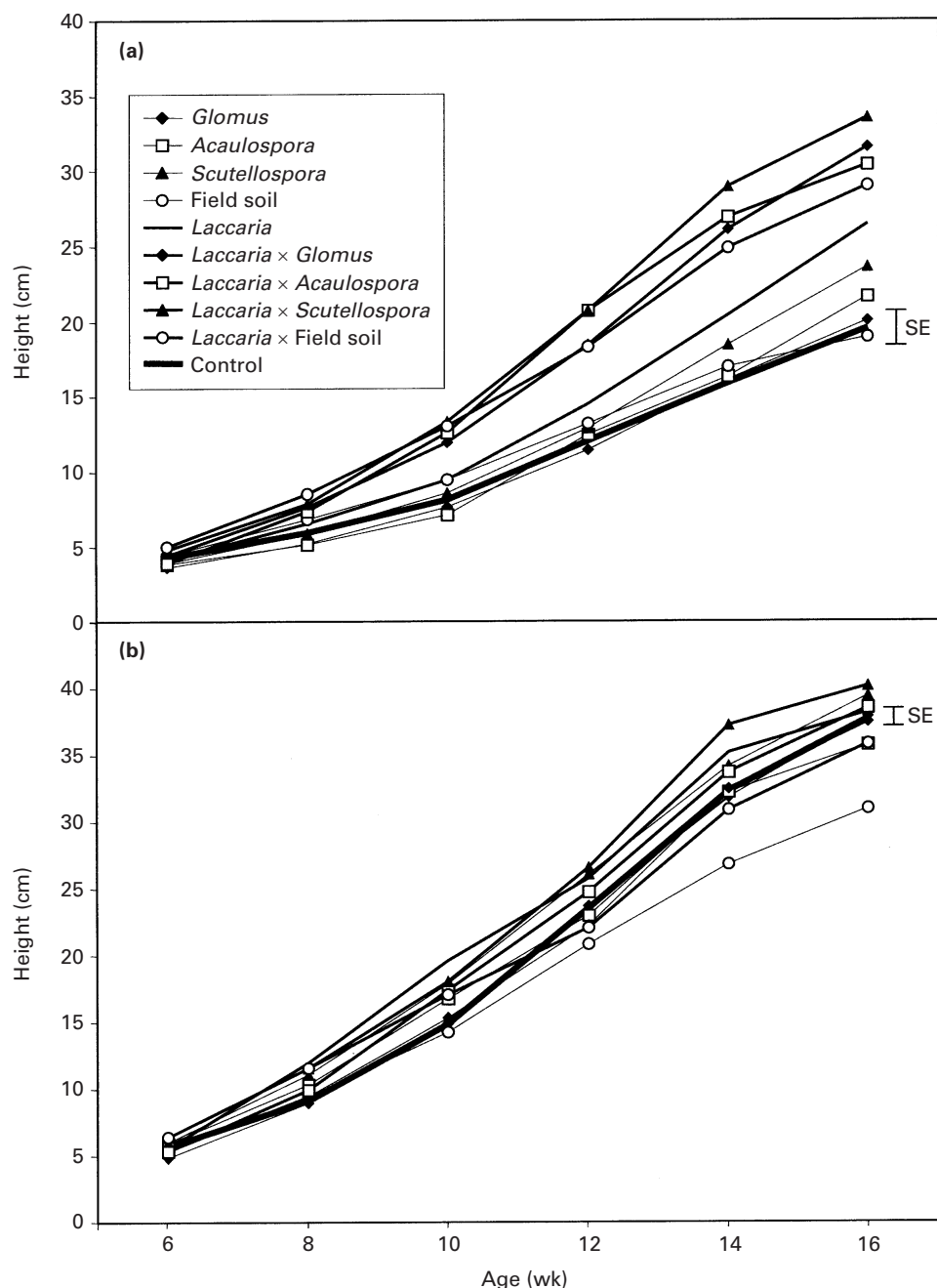


Fig. 3. Effect of different VAM and ECM inoculation treatments on changes in the height of *Eucalyptus globulus* seedlings over time. Results for two phosphorus concentrations are shown; 5 mg P kg⁻¹ soil (a) and 10 mg P kg⁻¹ soil (b). Treatments where the ECM fungus *Laccaria* has been inoculated, alone or with VAM fungi, are shown as thicker lines. Bars are two standard errors of the group mean for the final measurement (largest SE).

Eucalyptus species in this soil (N. Malajczuk *et al.*, unpublished). Nitrogen solution was applied every 3 wk (14 mg kg⁻¹ NH₄NO₃).

Soil was watered to field capacity (10% w/w) before planting and soil moisture levels were maintained by regular watering to weight. Four seedlings were planted in each pot, then thinned to two per pot after establishment. Pots were randomly rearranged weekly in a glasshouse on benches (*E. globulus*), or in root temperature tanks set at 25°C (*E. urophylla*). Air temperature was moderated by evaporative coolers and varied between 15–30°C.

Height measurements were taken every 2 wk starting at week 6. Root samples used to assess mycorrhiza formation were taken in two soil cores (1-cm diameter × 10-cm deep) from each pot using a stainless steel cylindrical corer sterilized with 70% ethanol at 8, 12 and 16 wk after planting. Only a small portion of the roots of each plant were sampled at each harvest. Cores were taken from different positions in pots at each harvest. The last cores were taken when plants were harvested at 16 wk. Root samples from cores were washed free of soil over a 100 µm mesh sieve. Roots were stored in 70%

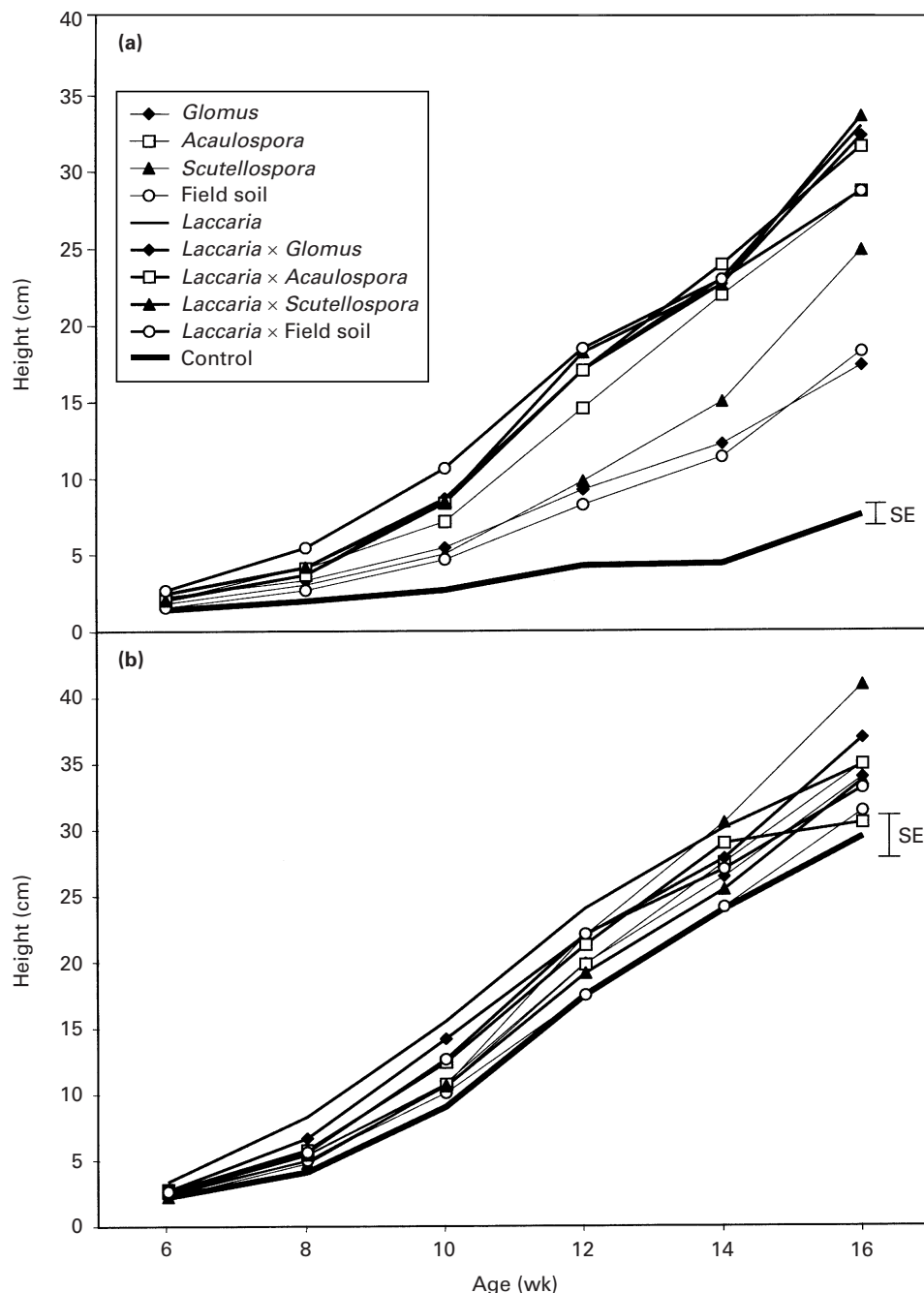


Fig. 4. Effect of different VAM and ECM inoculation treatments on changes in the height of *Eucalyptus urophylla* seedlings over time. Results for two phosphorus concentrations are shown; 5 mg P kg⁻¹ soil (a) and 10 mg P kg⁻¹ soil (b). Treatments where the ECM fungus *Laccaria* has been inoculated, alone or with VAM fungi, are shown as thicker lines. Bars are two standard errors of the group mean for the final measurement (largest SE).

ethanol, cleared with KOH and stained with trypan blue to quantify root length and mycorrhizal formation with a dissecting microscope (Brundrett *et al.*, 1996). Mounted slides of mycorrhizal roots were prepared and examined with a compound microscope to confirm the presence of mycorrhizal fungi. Seedlings were harvested after 16 wk by cutting off shoots and carefully washing soil from roots. Fresh and oven dry weights (30°C) of both shoots and roots were determined.

The growth and mycorrhizal colonization para-

meters measured were analysed by ANOVA for each combination of eucalypt species, sample time and P concentration. Changes over time were assessed by time-series analysis for each tree species. These data were also divided into fungus treatment categories (VAM only, ECM only, ECM and VAM treatments), before separate analysis to check for problems caused by the absence of variability in uninoculated treatments. Correlations between mycorrhizal colonization, specific root length and yield parameters were also examined.

RESULTS

Mycorrhizal formation

Soil coring was an efficient method of obtaining roots to quantify mycorrhizal associations with minimal damage to plant root systems. Periodic core sampling allowed a manageable number of pots to be used in the experiment and revealed changes in mycorrhizal colonization by different fungi over time.

When inoculated alone, all three isolates of VAM fungi colonized roots rapidly, occupying 50–90 % of seedling root length by 8 wk (Figs 1a,b, 2a,b). For most treatments, there were substantial reductions in the proportion of roots with VAM after 12–16 wk. The VAM fungi also colonized roots rapidly when inoculated with the ECM fungus *Laccaria lateritia*, but the proportion of roots with VAM declined to <10 % of root length after 12–16 wk in these treatments. Inoculation with VAM did not significantly affect ECM formation, but ECM inoculation significantly affected VAM formation (Table 1). Changes in ECM and VAM colonization and plant growth over time were highly significant (Table 1).

At later harvests, the decrease in degree of VAM colonization in plants with ECM and VAM was because most new roots formed ECM, since the total length of VAM roots did not decline (data not presented). There was a significant negative correlation between degrees of ECM and VAM colonization at 16 wk (Pearson correlation coefficient = -0.368 ; $P < 0.001$, $n = 160$). There also were significant ECM $\times$ VAM interactions in treatment effects on VAM formation and plant growth in many cases (Table 1).

Laccaria applied as spores was much slower to become established on the seedlings than VAM fungi applied as pot-culture inoculum (Figs 1, 2). There was a substantial increase in root colonization by *Laccaria* by 12 wk for *E. globulus* and by 16 wk for *E. urophylla*. These trends were similar when *Laccaria* was applied alone or with a VAM fungus (Figs 1c,d, 2c,d). These large increases in ECM colonization coincided with reductions in the proportion of roots with VAM, as already described. Additional evidence for substantial ECM fungus activity was provided by *Laccaria* fruiting in pots at 9 and 16 wk.

The overall effects of P fertilization on ECM or VAM formation by these eucalypts was not significant in most cases (Table 1). However, the higher P treatment appeared to reduce VAM colonization especially in the presence of *Laccaria* (Figs 1, 2).

VAM formation was slow in pots containing soil from a clover-based pasture prepared for plantation establishment (Figs 1, 2). This soil contained much lower degree of inoculum than soil from pot cultures. After 16 wk, there was ECM colonization in a few of

the pots inoculated only with VAM fungi or field soil (Figs 1, 2). This is consistent with our earlier observations of the time required for ECM fungus contamination to occur in glasshouse experiments (probably from airborne spores). It is unlikely that there was ECM inoculum in the field soil.

Growth responses

Differences in plant height owing to fungal treatments were apparent after 10 wk for both *E. globulus* and *E. urophylla* at the lower P but not at the higher P concentration (Figs 3, 4). Inoculation with *Laccaria* but not VAM significantly affected plant growth, especially at the lower P concentration (Table 1). *E. globulus* seedlings inoculated with *Laccaria* were larger than those inoculated only with VAM, but seedlings with VAM and *Laccaria* were generally larger than those just with *Laccaria* (Fig. 3a). For *E. urophylla*, all treatments with *Laccaria* were much larger than nonmycorrhizal controls, but those inoculated with VAM fungi were also substantially larger than the control (Fig. 4a). The isolates of VAM fungi varied in their effect on *E. urophylla* growth; *Acaulospora* produced the largest seedlings (similar in size to those inoculated with *Laccaria*), and seedlings with *Scutellospora*, *Glomus* or the fungi in field soil were intermediate in size relative to uninoculated seedlings (Fig. 4a). The differences in shoot dry weight owing to fungal treatments already described were similar to responses in plant height (Fig. 5a). There was a positive correlation between ECM colonization and shoot weight (correlation coefficient = 0.417 ; $P < 0.001$). There were also significant interactions between ECM and VAM inoculation treatments in their effects on plant growth (Table 1).

Mycorrhizal inoculation treatments had little effect on root:shoot biomass ratios (Fig. 5b). The presence of *Laccaria* substantially reduced the specific root length of both eucalypts relative to uninoculated controls or those with VAM only (Fig. 5c). Specific root length was inversely correlated with the number of ECM roots formed by plants at the final harvest (correlation coefficient = -0.514 ; $P < 0.001$).

DISCUSSION

When both types of fungi were present in pots, ECM replaced VAM as the dominant mycorrhiza type in *Eucalyptus* roots after several months of plant growth. This decrease in the proportion of roots with VAM apparently resulted from the preferential occupation of new roots by the ECM fungus *Laccaria*, since there was little change in the total length of VAM roots. Thus, after a slow start *Laccaria* became a more aggressive colonizer of roots of the two eucalypt species than any of the VAM fungi. The proportion of VAM in the eucalypt

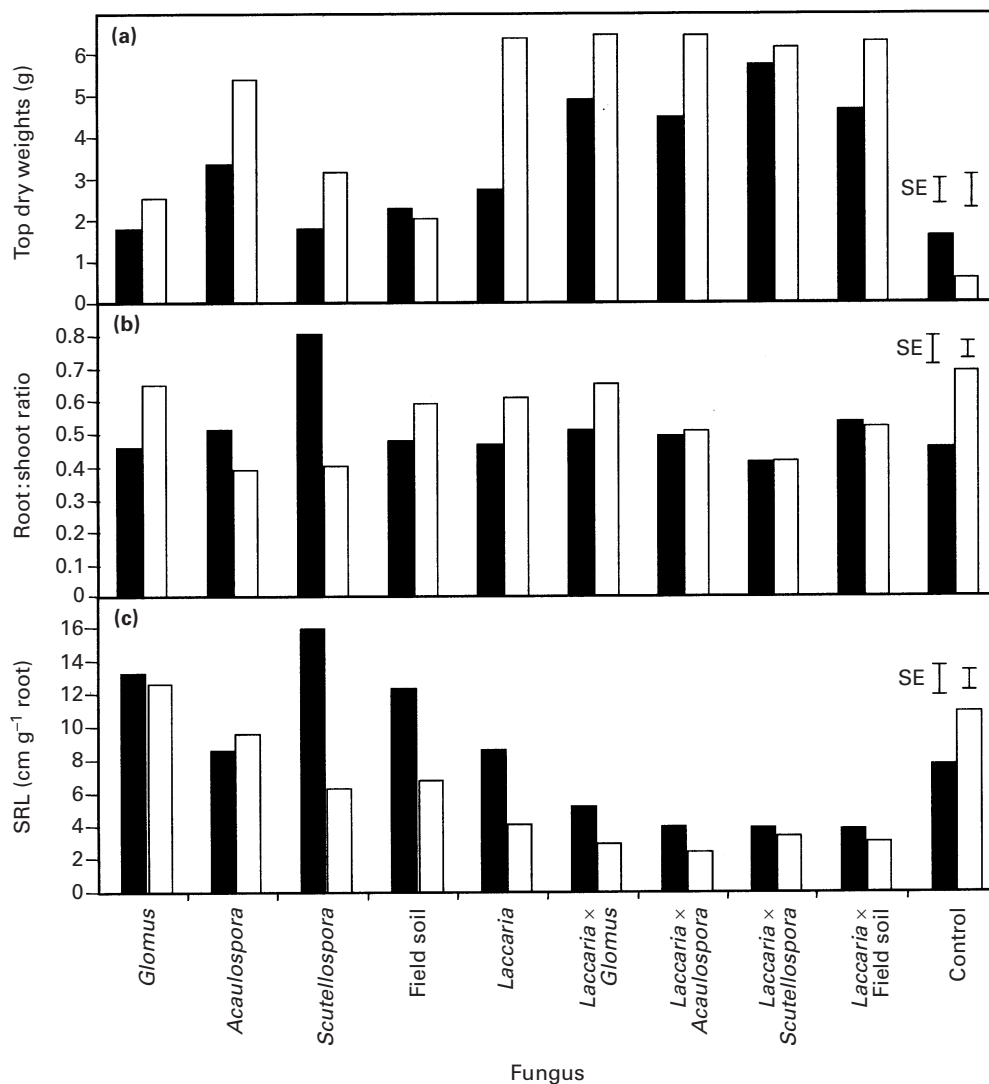


Fig. 5. Plant growth data for *Eucalyptus globulus* (closed bars) and *E. urophylla* (open bars) seedlings with different VAM and ECM inoculation treatments grown for 4 months at 5 mg P kg⁻¹ soil. (a) Shoot dry weights, (b) root:shoot biomass ratios, (c) specific root length (the ratio of root length:root biomass).

seedling roots also declined with time in the absence of ECM fungi and is believed to have resulted from changes in root properties (susceptibility to VAM fungi). Possible mechanisms for the observed succession from VAM to ECM dominance in eucalypt include:

- (1) changes in the relative capacities of different fungi to colonize roots in the soil may occur with time (inoculum potential);
- (2) the spread of one fungus in soil could be inhibited by antagonism from the hyphae of the other fungus;
- (3) ECM fungi on the root surface might prevent other fungi from gaining entry by acting as a physical barrier, or by some other means of antagonism;
- (4) changes to the quality or quantity of root exudates caused by a mycorrhizal association could influence the capacity of roots to attract other fungi;
- (5) mycorrhizal fungi might induce changes to roots that increase their resistance to subsequent fungal colonization;
- (6) changes to root morphology (a reduction in lateral root growth) because of ECM would probably reduce the efficiency of subsequent VAM colonization by reducing the available habitat;
- (7) changes to root growth and turnover rates may also be a factor – as mycorrhizas can only develop in new roots;
- (8) fungi that obtain more of a soil nutrient such as P, could adversely affect other fungi which obtain less, owing to a linkage between plant carbohydrate and nutrient exchange;
- (9) increased carbon supply to one fungus might reduce quantities available to others, even if they occupy separate roots.

The capacity for ECM fungi to replace VAM fungi might result from differences in their growth

strategies. One likely explanation is that ECM fungi maintain a higher inoculum potential than VAM fungi in soils. This suggests that relative to VAM fungi, ECM fungi can (1) produce more hyphae in soil, (2) concentrate more hyphae near growing roots, (3) have more hyphae on the surface of roots in close proximity to emerging laterals, (4) respond more rapidly to root initiation, or (5) redistributes resources to sites of root growth more efficiently. It is generally assumed that ECM associations obtain more energy from host plants than VAM associations (Smith & Read, 1997) and this could result in a greater capacity for hyphal activity.

It has been suggested that the outcome of competition between mycorrhizal fungi of the same or different types could be mediated by chemicals of fungal or host origin, mechanical barriers, competition for root carbohydrates, effects on rhizosphere communities, or other factors (Lodge & Wentworth, 1990). The supply of carbon in roots which is available for mycorrhizal fungi is likely to be limited, and thus carbon use by one fungus may affect other fungi. For example, the presence of one VAM fungus can inhibit mycorrhizal formation by another fungus, even when they occur on separate parts of the root system (Pearson *et al.*, 1993). Once ECM fungi are established, they might spread rapidly to new root initials, perhaps preventing subsequent entrance by VAM fungi if the mycelial sheath formed by ECM fungi functions as a barrier (Chilvers *et al.*, 1987). However, the effect ECM fungi have on specific root length might have a greater impact on VAM fungi by reducing the production of fine roots they can colonize.

Spatial heterogeneity in fungus inoculum adds another dimension to competitive interactions in soils in natural ecosystems where ECM and VAM fungi occupy separate soil domains (Brundrett & Abbott, 1995). Hyphae of some ECM fungi are known to have the capacity to become locally dominant in soils (producing mats and strands). For wetland tree species, soil moisture can regulate the balance between ECM and VAM in roots, as waterlogging favours VAM, but ECM become dominant when soil dries (Lodge, 1989; Khan, 1993).

We think that VAM were more successful than *Laccaria* in colonizing eucalypt seedling roots during the first 2 months because there were higher initial inoculum levels of the VAM fungi and they were more uniformly distributed in the soil. In competition between different VAM fungi, both the amount and distribution of fungal inoculum is considered to be important, as fungi whose propagules are more abundant or more uniformly distributed through the soil are normally at a considerable advantage (Wilson, 1984; Hepper *et al.*, 1988). *Laccaria* required several months to become established from spores. The concentration of spores

and the method of application was optimized during earlier glasshouse experiments, which found spore inoculation to be as effective as other methods (Lu *et al.*, 1998). The slow establishment of *Laccaria* may have resulted from delayed spore germination and/or the time required for hyphae to spread from the inoculation point. Initially, there would probably have been a much greater biomass of VAM fungi than ECM fungi, but the relative number of propagules is not known. In experiments with ECM fungi there is typically a rapid increase in the rate of root colonization after a slow start. This is attributed to secondary infection from existing mycorrhizal roots (Chilvers & Gust, 1982).

Mycorrhizal formation by VAM fungi declined even in the absence of *Laccaria* after several months providing evidence that eucalypt roots became less susceptible to these fungi with time. However, the most severe reductions in degree of VAM colonization occurred when *Laccaria* was present, so competition had the greatest impact on VAM fungi. The isolates of VAM fungi used in this experiment continued to form mycorrhizas for 6 months or longer, in past experiments with other hosts. Thus, it is unlikely that changes to fungal activity initiated a decline in VAM formation. Succession from VAM to ECM associations as *Eucalyptus* seedlings age also occurs in the field (Lapeyrie & Chilvers, 1985; Bellei *et al.*, 1992). It would be unwise to assume that ECM fungi will always outcompete VAM fungi in eucalypt roots, as experiments have included only a few isolates of ECM and VAM fungi or species of *Eucalyptus*. However, the available information suggests that difference between individual ECM or VAM fungi are small relative to functional differences between the two types.

Future studies of interactions between competing mycorrhizal fungi or associations should consider changes in their relative importance over time, as measurements of fungal activity or plant responses at a single time can be misleading. Unfortunately, many mycorrhizal studies have measured plant growth and mycorrhizal formation only once after an arbitrary period of growth. It is also important to use well characterized fungal isolates, which have been proven to be compatible with the host species, and attempt to standardize inoculum levels, since differences in inoculum potential between fungi can overwhelm short-term differences in their competitive ability. In experiments with *Eucalyptus* species, soil nutrient concentrations must be carefully chosen, as responses to mycorrhizas are only likely to occur in very infertile soils. Plant genotypes also need to be considered, as some *Eucalyptus* species are known to obtain large benefits from VAM fungi (Jasper & Davy, 1993), but most others probably do not.

The occurrence of two mycorrhizal types in the same root system raises important questions about

the relative benefits they provide to the host and mechanisms of competition between the two types of fungi. We do not know whether the competition which occurs between ECM and VAM fungi is mediated internally by plants through root C supply or other means, by habitat-space limitation within roots, or by interactions between hyphae in soil. We also know little about competition for soil nutrients between hyphae of the two associations, or their capacity to access different forms of these nutrients. These topics provide many future research opportunities.

This study and most earlier experiments have shown that ECM associations are usually more important than VAM associations for *Eucalyptus* species. However, there is evidence that VAM associations can provide benefits to eucalypts, especially during seedling establishment. There are also cases where the benefits provided by VAM and ECM together can exceed those provided by either one alone. The importance of VAM associations of eucalypts is likely to be greater in disturbed habitats or exotic locations where there are few eucalypt-compatible ECM fungi (Lu *et al.*, 1999), since VAM fungi generally exhibit little host specificity.

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